Emulsion

User guide for Emulsion Lab and Emulsion Lite

Emulsion Lab

The desktop editor for Windows. One-time purchase, no subscription: RAW, AI tools, the Library and full-size export.

Emulsion Lite

The free web edition. The same editor in your browser, on a computer or a phone, with a few limits. lite.emulsionlab.io

Version **2.6.1**

Your photos stay on your device.

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How to read this guide. Everything applies to both editions unless it is marked LAB. In Emulsion Lite, Lab's tools are still visible but greyed out with an "Included in Emulsion Lab" link, so you always know where they are.

1 Getting started

Emulsion is a photo editor you buy once and keep. There is no account and no subscription, and your photos never leave your device.

Emulsion Lab on Windows LAB

1. Run the installer, **Emulsion Lab_2.6.1_x64-setup.exe**. It installs for your Windows user only, so you don't need administrator rights.
2. If Windows shows "Windows protected your PC", click **More info** and then **Run anyway**.
3. Emulsion Lab needs Microsoft WebView2. Windows 10 and 11 already include it, and the installer fetches it if it is missing.
4. Start Emulsion Lab. The first time, it asks for your **license key** (from your purchase email). Paste it and click **Activate**. This is the only time Lab needs the internet.

> **After activation Lab works fully offline.** It never checks the key again and never stops working because you are offline. It makes only two network calls, and only when you click: Activate, and Deactivate.
>
> To move Lab to another computer, open **About > License** (click the logo) and click **Deactivate this computer**. That frees the activation so you can use the key on the new machine.

Updates install over the old version and keep your license, Library and settings.

The start-up screen with the Emulsion Lab logo and the credits appears only on the first start after you install or update Lab. After that Lab opens straight to your work. The credits and licences are always in **About** (click the logo).

Emulsion Lite in the browser LITE

1. Open https://lite.emulsionlab.io in a current browser: Chrome, Edge, Firefox or Safari, on a computer, tablet or phone.
2. Click **Open a photo**, drop a photo on the page, or paste one. You can also click **Try a sample**.

Lite runs entirely in your browser. It does not upload your photo or contact any other server. The only file it loads after the page is its own pack of built-in presets, the first time you open the Presets tab.

Your first edit in five steps

1. Open a photo (Ctrl + O).
2. Go to the **Adjust** tab and click **Light**. Try **Auto tone**, then set Exposure and Contrast to taste.
3. Open the **Presets** tab and hover over a preset to preview it. Click it to apply it.
4. Hold \ to compare with the original.
5. Click **Export** and save a JPEG.

Everything is undoable. Ctrl + Z undoes and Ctrl + Shift + Z redoes. The **History** tab lists every step by name; click one to jump back to it.

2 Opening photos

You can open, drop or paste photos, and work on several at once.

Ways to open a photo

- **Open** in the header, or Ctrl + O. This also opens Emulsion projects (**.emulsion**).
- Drag one or more photos onto the canvas.
- Paste an image (Ctrl + V). With a photo already open, the pasted image becomes a new layer.
- **New** makes a blank canvas. It has size presets (Instagram, Story, 4K, A4 and more) and a white, black, transparent or colour background. The shortcut is Ctrl + N in Lab and Ctrl + Alt + N in Lite.
- LAB Double-click a **.emulsion** project in Explorer, or use **Open with**. The start screen lists recent files with thumbnails.

Several photos at once

Drop or open several photos and Emulsion asks how to open them:

- **Separate tabs**: each photo opens in its own tab above the canvas. Switch tabs with Ctrl + Tab and go back with Ctrl + Shift + Tab. Each tab keeps its own layers and undo history.
- **As layers**: the photos are stacked as layers of one document.

Up to 12 photos can be open at once on a computer, 4 on a touch device and 3 on iPhone and iPad. Each open photo uses memory. In Lab, Ctrl + W closes the current photo.

File types

Type	Lite	Lab
JPEG, PNG, WebP and other images your browser can read (colour profiles applied, see Colour and export)	✓	✓
TIFF, 8 and 16 bit (uncompressed, LZW, ZIP or PackBits; grey or RGB, with transparency), colour profile applied	–	✓
Emulsion projects (.emulsion)	✓	✓
Lightroom / Camera Raw presets (.xmp), dropped or imported	✓	✓
LUTs (.cube) for the Look adjustment	✓	✓
Camera RAW: Panasonic RW2, Sony ARW, Canon CR3/CR2, Nikon NEF, Fujifilm RAF, Hasselblad 3FR/FFF, DNG, Olympus ORF, Pentax PEF and more	–	✓

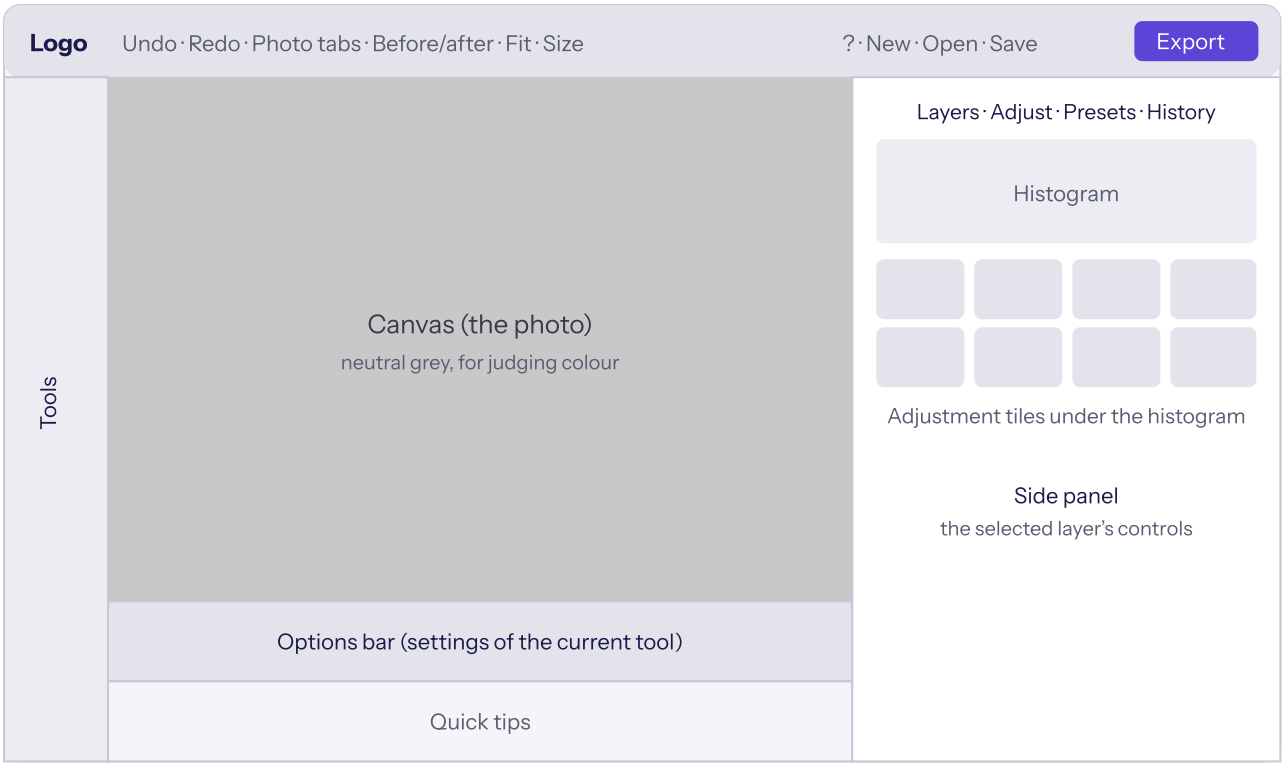
Photo size limits

Lite reduces very large photos when it opens them: to 36 megapixels on most devices, and to about 16 megapixels on iPhone and iPad, where browsers limit image size. **Lab** opens photos up to 250 megapixels at full size.

Memory. Before a photo opens, Emulsion checks that it fits in the memory this computer has left: about 4 GB of an 8 GB computer, three quarters of a bigger one. One that will not fit is not opened, and a message says how much it needs and how much is left, so the window never runs out in the middle of opening it. A 24-megapixel RAW needs about 1 GB while it opens and 0.3 GB while it stays open; a 45-megapixel one about 1.8 and 0.5 GB. Close a photo (or another program) and open it again. When several photos open at once, the ones that fit open and the message names the first that did not.

3 The workspace

Emulsion keeps every tool in view, not inside menus.



On a computer: the header at the top, tools on the left, the canvas in the middle, the side panel on the right.

The parts

- **Header**: Undo and Redo, tabs for open photos, **Before/after**, **Fit**, the photo's size in pixels, **?** (all shortcuts), **New**, **Open**, **Save** and **Export**. Click the logo for About and licences. LAB also has a **Library / Edit** switch here.
- **Tools**: Move, Pan, Select, Ellipse, Lasso, Wand, Object, Crop, Brush, Eraser, Clone, Remove, Heal, Liquify, Dodge, Gradient and Picker. Hover over a tool to see its key.
- **Options bar**: the settings of the current tool, such as brush size, selection mode, crop ratio or feather.
- **Quick tips**: the most useful shortcuts for what you are doing right now. Close the bar with ×, and turn it back on from the shortcut sheet (?).
- **Side panel**: four tabs. **Layers** (the layer stack and the selected layer's blend, opacity and mask), **Adjust** (the histogram, adjustment tiles and the controls of the selected adjustment), **Presets** and **History**.

Viewing the photo

- Zoom with the scroll wheel, and pan with Space+drag or the Pan tool (H). On touch screens, pinch to zoom and drag with two fingers to pan.
- **Fit** shows the whole photo. The zoom level and the photo's pixel size are shown in the header.

- **Before and after**: hold \ to see the original, or click **Before/after** for a split view with a divider you can drag.
- **Clipping warnings**: press K or click the small triangles on the histogram. Blown highlights show in red and crushed shadows in blue.

On a phone LITE

The canvas is at the top, then the tool strip, then the panel. Swipe the tool strip sideways to reach more tools, and fold the panel away with the arrow next to its tabs. Buttons are sized for fingers.

4 Adjustments

Every adjustment is a layer. You can change it, hide it, mask it or delete it later without touching the photo's pixels.

Adding an adjustment

1. Open the **Adjust** tab.
2. Click a tile under the histogram, such as **Light** or **Curves**. A new adjustment layer appears above the selected layer.
3. Drag its sliders. Every slider also takes a typed value: click the number, type, and press Enter (Esc reverts; the arrow keys nudge).

> **Find an adjustment by typing.** Press Ctrl+F and type a word such as "grain" or "white balance". Enter adds the matching adjustment and jumps to that setting.

To change an adjustment later, click its layer in the **Layers** list. Its controls open right there.

Limiting an adjustment

- **Mask it**: paint on the adjustment layer with the Brush or Eraser and it gets a mask automatically, or use a Gradient (G). With the photo selected, a gradient makes its own adjustment layer: drag it, then move the sliders. See chapter 7.
- **Limit to a brightness band or a colour**: in an adjustment's **Limit to** control choose **Brightness** (for example, only the shadows) or **Color** (pick a colour on the photo).
- **Limit to a selection**: make a selection first. See Adjusting only inside a selection.

The adjustments one by one

Tile	What it does
Light	Exposure, contrast, highlights, shadows, white balance and colour in one place. **Auto tone** and **Auto white balance** are here.
Curves	A tone curve for RGB or each colour channel, with point editing and region sliders.
Mixer	Hue, saturation and brightness for 8 colour ranges. **Pick on photo**: drag on a colour in the photo to change it.
Grade	Tint the shadows, midtones and highlights separately.
B&W	Black and white with colour filters.
Look	Apply a LUT from a **.cube** file, one of My LUTs, or one of 6 built-in looks.
Detail	Sharpening, clarity (local contrast) and Dehaze.
Noise	Noise reduction for luminance, detail and colour. It preserves edges.
Lens	Removes colour fringes (chromatic aberration) and dark corners (vignetting).

Tile	What it does
Blur	Softens the image. Add a mask to blur only part of it.
Motion	Panning blur along a direction. Keep the subject sharp with a mask, which is ideal for motorsport.
Spin	Spinning wheels. Tap a wheel to place an ellipse, then set its size, shape and tilt. Use one layer per wheel.
Effects	Vignette and film grain.
Halation LAB	A glow around bright lights, as film shows it: a red-orange fringe close to the light and a wider haze. **Strength**; **Radius** (in percent of the photo's long side, so it looks the same at any size); **Threshold** (how bright a light must be to glow); **Warmth** (0 is the light's own colour). The glow is added like light, so it never goes past white, and parts of the photo far from any light do not change. Mask it to keep it off some lights. Halation is an Emulsion Lab feature: in Lite the tile is greyed, and a Lab project with a Halation layer opens there with the layer kept (not shown), so it is still there when you open the project in Lab again.
Skin LAB	Smooth skin. Evens out skin tone and blemishes but keeps pores and edges. It starts hidden: paint it onto the skin with a soft brush.
Depth LAB	Lens blur. Blurs the background (and foreground) by distance like a wide aperture, using an on-device depth map. Pick the focus point on the photo.

Auto tone and Auto white balance

The **Auto** buttons in **Light** look at the photo under the layer, only where the layer changes it: through its mask or brightness / colour limit, or inside the selection when you use the Adjust tab with a selection. If that area is under about 1% of the photo, Auto leaves the sliders alone and says so.

- **Auto tone** sets Exposure, Contrast, Highlights, Shadows, Whites and Blacks. It keeps the mood of the photo: a normal scene comes most of the way to a medium brightness, a bright one stays bright, and a dark one (a night race, a dim pit garage) stays dark and changes very little. A short message says when Auto held back.
- It lifts exposure by at most 1 stop, and less as the ISO rises (a quarter stop from ISO 3200), because lifting a noisy photo shows its noise. It never lifts the blacks, and it does not lift the shadows of a high-ISO photo.
- It protects highlights: when lights in a dim scene (headlights, floodlights) are close to white it adds no exposure, and it never makes more than a small part of the photo, or of its lights, clip to white.
- The ISO comes from the photo's camera info (EXIF). A photo without it is treated as ISO 800.
- **Auto white balance** is separate and only corrects a clear colour cast, by at most ±25 Temperature and ±15 Tint. A photo whose colour is already right stays as shot: a sunny scene, green grass or warm track lights are not a cast. On night and stage scenes it also leaves the colours as shot, since their colour is the light's. Auto tone never changes the white balance.

> **Preview and full size.** To stay fast, the editor previews at a reduced size (about 4 megapixels on a computer, 2 on touch devices). Export always renders at full quality. Sharpening, blur and grain can look slightly different at 100% than when the photo is fitted to the screen.

My LUTs

Import your **.cube** LUTs once and pick them from a list afterwards, in both editions. My LUTs is in the Look panel, under the built-in looks.

- **Import**: **Import .cube…** takes one or several files, or drop them onto the My LUTs block. Each one is copied into Emulsion's own storage (Lab: the app's data folder; Lite: this browser), so it is listed every time you start, even if the original file moves. After **Load .cube file…** the panel offers **Add to My LUTs** for that file.
- Each LUT shows its name, its size (**33³**, **65³**, or **1D 1024**) and a small preview on your photo. **One click** puts it on the Look layer, as one undo step. Search by name; ⋯ renames or deletes it (Lab also has **Save .cube…**, which writes the file back exactly as it was imported).
- The same LUT is never imported twice, even under another name: Emulsion compares the table itself. A different LUT with a name already taken gets a number, as in **Teal (2)**.
- Supported: 3D LUTs from 2 to 65 points per side, 1D LUTs up to 65,536 entries, with **DOMAIN_MIN** / **DOMAIN_MAX** and a **TITLE**. A file over 32 MB, a damaged or incomplete one, or a LUT larger than 65 points is refused with a message saying why. My LUTs holds up to 300 LUTs and 512 MB.
- **Projects stay complete**: a Look layer keeps its own copy of the LUT, as before, so renaming or deleting a LUT in My LUTs never changes a saved project or a photo you are editing.
- **Lab**: **Open LUT folder** shows My LUTs' own files in Explorer. **Watch a folder…** lists every .cube in one folder of yours (marked **folder**, read only: rename or delete them in the folder itself). The list is read again when the panel opens, when you come back to Emulsion, and with **Refresh**.
- To turn your own edits into a LUT, use **Export look (.cube)** in the Presets tab (see Presets). It is close to the edit but not exact, and only colour adjustments without a mask go into it, so it is not offered as a My LUTs entry.

Crop, straighten, geometry and size

The Crop tool (C) has four modes in the options bar:

- **Crop**: drag the handles, then press Enter to apply or Esc to cancel. Ratios include 1:1, 4:3, 5:4, 7:5, 3:2, 16:9, 16:10, 21:9, cinema ratios and the XPan's 65:24. Press X to swap portrait and landscape. The crop's size in pixels shows on the box. Straighten with the slider or **Level line**, and rotate in 90° steps. Guides: thirds, golden ratio, golden spiral, diagonals, triangle and grid.
- **Geometry**: vertical and horizontal perspective, lens distortion and rotation, plus **Auto level** and **Auto upright**.
- **Size**: resizes the image (high-quality Lanczos resampling of every layer and mask).
- **Canvas**: adds or trims space around the photo, with an anchor and a fill colour. **Pad to ratio** adds a border to reach a ratio.

Each of these is one undo step.

5 Selections

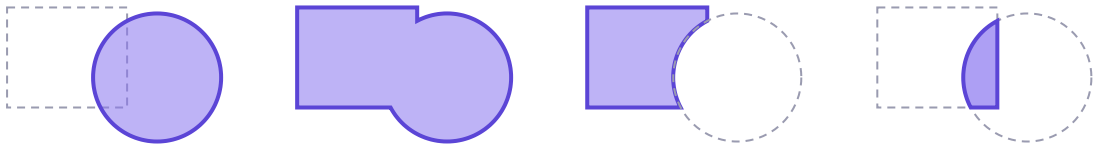
A selection limits what you paint, fill, delete, remove or adjust to one part of the photo.

Selection tools

Tool	Key	Use
Select (rectangle)	M	Drag a rectangle. Ctrl + A selects everything.
Ellipse		Drag an ellipse.
Lasso	L	Draw freehand around an area.
Magic wand	W	Click to select similar colours. Raise **Tolerance** to select more.
Object	O	Drag a box around an object and Emulsion finds its edges. A snug box gives cleaner edges. Then use **Paint +** and **Paint −** to add missing parts or remove extra ones. LAB traces the object with on-device AI and adds **Select subject**; Lite uses colour.

Selection modes

Every selection tool can make a new selection or combine with the one you have. Pick a mode in the options bar, or hold a key while you drag:



The rectangle is the selection you had, the circle the one you draw next. The filled area is the result.

Mode	Hold while dragging	Result
New		Replaces the selection.
Add	Shift	Adds the new area to the selection.
Subtract	Alt	Removes the new area from the selection.
Intersect	Shift + Alt	Keeps only where the two overlap.

The cursor shows the current mode. On a touch screen, tap **Add**, **Subtract** or **Intersect** first. **Paint +** and **Paint −** refine any selection with a brush. Every selection change is one undo step, and an undone stroke stays undone: the next stroke starts from the selection as it is now.

Working with a selection

- **Move the pixels**: drag inside the selection. The pixels lift onto a new layer. Turn on **Fill gap** to fill the hole left behind with content-aware fill, and use **Feather** to soften the edge.
- **Copy**: Alt +drag with the Move tool (V) moves a copy. Ctrl+C or Ctrl+J copies to a new layer, and Ctrl+X cuts to a new layer.
- **Delete** clears the selected pixels. Shift+Delete removes them with content-aware fill.
- **Invert** with Ctrl+Shift+I, and **deselect** with Ctrl+D or Esc.
- **Mask from selection**: in the Layers tab, **From selection** turns the selection into the layer's mask.

Adjusting only inside a selection

With a selection active, the Adjust tab shows *"Applies to selection only. Deselect for the whole photo."*

1. Make a selection, for example the sky with the Magic wand.
2. In the **Adjust** tab, click a tile or change a slider.
3. The change goes onto an adjustment layer masked to the selection, with soft edges from **Feather**. The layer shows a small **Selection** badge in the Layers list.

Emulsion keeps your other adjustments safe:

- If you change an adjustment that covers the whole photo while a selection is active, your edit goes onto a **new** layer of the same kind, masked to the selection. The original layer is untouched.
- If you change a selection-limited adjustment with no selection active, the edit goes onto a new layer for the whole photo.
- Either way it is one undo step, and a message tells you a new layer was made.
- To change a layer exactly as it is, click it in the **Layers** list and edit it there.

Presets always apply to the whole photo, even with a selection active.

6 Transform

Resize, rotate and flip the selected pixels with a box and handles.

Transforming pixels

1. Make a selection around the pixels to transform (Ctrl + A for the whole layer). The Transform buttons appear in the options bar.
2. Click **Transform**, or press Ctrl + T in Lab or Ctrl + Alt + T in Lite (the browser keeps Ctrl + T for new tabs).
3. Drag the eight handles to resize, drag just outside a corner to rotate, and drag inside to move.
4. Press Enter to apply or Esc to cancel.

While dragging	Effect
Shift on a handle	Keeps the shape (proportions).
Alt on a handle	Scales from the centre.
Shift while rotating	Snaps to 15° steps.
Options bar	Type exact width %, height % and angle.

The pixels are resampled once, in high quality, when you press Enter, and the whole transform is one undo step. Switching tools, saving or exporting applies a transform in progress first.

Transform selection

Transform selection changes only the selection's outline, not the pixels. Use it to fine-tune a selection's size or angle before you adjust or copy it.

Layer transforms

A raster layer can also be scaled, rotated and flipped without changing its pixels: select it and use the controls in the **Adjust** tab. Move a layer with the Move tool (V).

> **Masks stay in place.** A mask belongs to the document, not the layer, so it does not move when you move or transform its layer.

7 Layers and masks

Your photo, your paint and every adjustment are layers in a stack. The top of the list is the top of the stack.

The Layers tab

- Click a row to select that layer. Click the eye to hide or show it.
- **Select several layers**: Ctrl +click a row to add it to the selection or take it out, Shift +click to select every row from the last one you clicked. Selected rows are tinted. A plain click selects one layer again. Dragging a row still moves just that row.
- **+ Photo** adds another photo as a layer. **+ Paint** adds an empty layer to paint on.
- The buttons under the list move the selected layer up or down, duplicate it, **merge visible** layers into a new layer, **merge selected** layers, or delete it.
- **Merge selected** (Ctrl + E, with two or more layers selected) turns the selected layers into one layer, with their opacity, blend modes and masks. It takes the place and the name of the topmost one. Hidden layers in the selection are left as they are. An adjustment layer in the selection is baked into the selected layers under it only; one with no selected layer under it can't be merged. Layers of an applied preset can't be merged (use Merge visible). If other layers sit between or under the selected ones, the result can look a little different, and Emulsion says so the first time. One undo puts all the layers back.
- **Blend** offers 17 blend modes: Normal, Multiply, Screen, Overlay, Soft light, Hard light, Color dodge, Color burn, Darken, Lighten, Difference, Exclusion, Hue, Saturation, Color, Luminosity and Linear light. There is also an **Opacity** slider.
- **Copy edits** / **Paste edits** (Ctrl + Shift + C / V) copy this photo's adjustments to another open photo.

Masks

A mask shows or hides parts of a layer without erasing anything. In the Layers tab, under **Mask**:

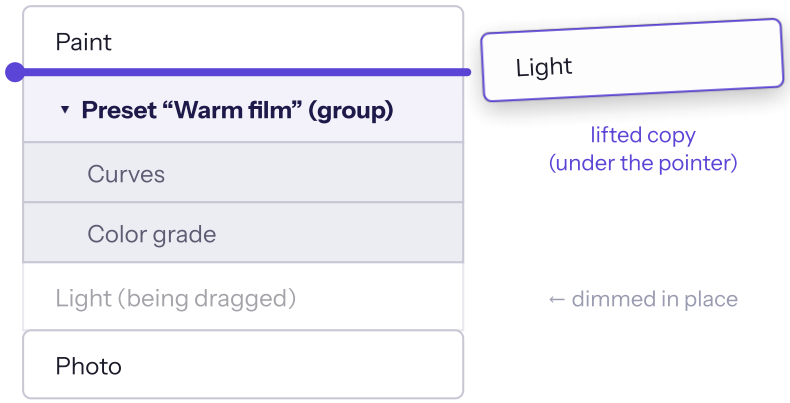
- **Add mask** (everything shown), **Hide all** (everything hidden), or **From selection**.
- With the mask selected, the **Brush reveals**, the **Eraser hides** and the **Gradient fades**. On a photo layer, switch between **Edit photo** and **Edit mask**, or click the mask thumbnail.
- **Mask on/off**, **Invert** and **Delete**.
- On an adjustment layer you can simply paint: the first stroke creates the mask.
- **Gradient** (G) draws an editable fade. Its handles stay adjustable afterwards. Drawn on an adjustment layer, it becomes that layer's mask. Drawn with the photo selected, it adds a new **Linear gradient** or **Radial gradient** layer (Light & color) whose sliders change the photo only through the gradient; to fade the photo itself, choose **Edit mask** first. Radial gradients have **Feather**, and **Flip** applies the effect outside instead of inside. **Golden hour** is a ready-made warm gradient.
- LAB **Subject**, **Background** and **Sky** make a mask with on-device AI (chapter 9).

A small badge on a row shows when an adjustment is limited: **Selection**, **Gradient** or **Masked**.

Preset groups

An applied preset is made of several adjustment layers. The Layers list shows it as **one group row** with an eye, an expand arrow, **Reset to preset** and a remove button. Click the row, or the arrow, to open it and see its layers. Click a layer inside to edit it. In Lite, built-in presets' layers are locked (chapter 8).

Reordering layers (drag)



The dragged row dims in place, a lifted copy follows the pointer, and a line shows where it will land.

- **Mouse**: press on a row and move it up or down. A plain click still just selects the row.
- **Touch**: touch a row and hold it for a moment, then drag. If you move right away, the panel scrolls instead.
- Near the top or bottom of the panel, the list scrolls by itself.
- Press Esc to cancel a drag. Letting go where the row started changes nothing.
- One drag is one undo step, and the selected layer stays selected.

Rules for preset groups:

- Dragging a group row moves the whole preset.
- A layer inside a preset can only be reordered within that preset.
- Other layers can't be dropped inside a preset.
- In Lite, the layers of a built-in preset can't be reordered, but the group row can still move.

The up and down buttons under the list still work. They move the selected layer one step and step over a whole preset at once.

Painting and retouching tools

Tool	Key	Use
Brush / Eraser	B / E	Paint or erase on a layer or its mask. [and] change the size.
Picker	I	Pick a colour from the photo for the brush.

Tool	Key	Use
Clone	S	Alt+click sets the source (on touch, the first tap), then paint.
Remove	R	Paint over something to remove it. Emulsion fills it from its surroundings. LAB adds AI fill.
Heal LAB	J	Paint over a blemish or hair. / tries another source and Alt+click picks one. Fixes go on a separate Retouch layer. **Visualize spots** helps you find dust.
Dodge & burn LAB	D	Paint light (Dodge) or dark (Burn) on a separate layer. Hold Alt to switch.
Liquify LAB	Q	Push, Bloat and Pinch on the photo layer. Each stroke is one undo step.
Frequency separation LAB		In the Layers tab, it splits the photo into Low (tones) and High (texture) layers for skin work.

8 Presets

A preset is a complete look in one click. You can stack several and dial each one up or down.

Where presets come from

- **Built-in presets**: 431 looks ship with Emulsion (from the MIT-licensed Lightroom-Presets collection by peva3, renamed). Lab has them at first launch. Lite loads them the first time you open the Presets tab.
- **Your own**: **Save as preset** keeps your current adjustment layers as a preset. Gradient masks scale to each photo.
- **Imported**: Lightroom / Camera Raw **.xmp** presets. Import them in the Presets tab, or drop them on the window. Common sliders and Look tables are imported. Texture and the parametric curve are not.

Applying presets

1. Open the **Presets** tab. Your presets come first, then the built-in groups. Search by name at the top.
2. Each preset shows a thumbnail of your photo. Hover to preview it on the canvas.
3. Click to apply. Each new preset goes **on top** of the ones already applied, so it builds on them.

Clicking a preset that is already applied asks whether to **Apply again**, **Replace** or **Remove**.

The Applied list and Amount

At the top of the Presets tab, the **Applied** list shows the stack. Each preset has:

- an **Amount** slider that mixes the preset with the picture before it (100% is the full preset)
- a remove button

Remove all clears the stack. Applying, removing and changing Amount are each one undo step.

Editing a preset's layers LAB

In Lab, every applied preset can be changed. Expand its group in the Layers list, click one of its layers, and edit it like any other adjustment: sliders, mask, opacity and blend.

Reset to preset

When you have changed a preset's layers, **Reset to preset** (the undo-arrow button on the group row, also in the Applied list) puts them back exactly as the preset applied them. Amount, visibility and masks stay as they are. It is one undo step.

Presets in Lite LITE

- **All 431 built-in presets** are included and applied as they are. Their layers are locked: you can show or hide a preset, set its Amount and remove it, but not change its sliders or reorder its layers.

- Up to **2 presets** can be stacked at a time.
- Your own presets and imported **.xmp** presets can be edited in Lite as in Lab.

Managing the list

- **Delete** your own or imported presets. Emulsion asks first. Built-in presets can't be deleted, but they can be **hidden**. **Show hidden presets** and **Restore all hidden** bring them back.
- **Select presets** mode, Ctrl/Shift+click, and the right-click menu (long-press on touch) act on several at once.
- **Export the look as a .cube** LUT, for use in a camera or video software. It includes only colour adjustments that apply to the whole picture without a mask.

9 AI tools LAB

Emulsion Lab's AI runs entirely on your computer. Nothing is uploaded.

The AI tools use your graphics card through WebGPU when it is available (AMD, Intel and NVIDIA alike), and otherwise the processor. If a model fails on the graphics card, Lab runs it on the processor instead. That is slower but gives the same result. In Lite these tools are shown greyed out.

AI masks

In the Layers tab, under **Mask**: **Subject**, **Background** and **Sky** build a mask for the selected layer. They replace any existing mask. Refine the result with the Brush and Eraser.

Select subject and AI object tracing

- **Select subject** (Object tool) selects the main subject in one click.
- With the Object tool's **AI** switch on, a box you drag is traced by AI. **Paint +** and **Paint −** strokes correct it only near where you paint: + only adds, − only removes.

AI fill

The Remove tool can fill with AI, which works well for larger objects. **Patch** keeps the classic fill.

Lens blur (Depth)

The **Depth** tile estimates how far away each part of the photo is, then blurs by distance. Click the photo to set the focus, then set the depth of field and the foreground blur. The depth map is saved in the project.

AI denoise

1. Select a photo layer and click **AI denoise** in the Layers tab.
2. A layer named "AI denoise" appears above the photo. A progress panel with **Cancel** shows on the canvas while it works.
3. **Strength** blends it with the original. Clicking again removes it, and **Denoise again** reruns it.

Full-size RAW files are processed in tiles, so they fit in memory. A preset applied afterwards lands above the denoised layer.

With a graphics card it takes about 12 seconds for a 24-megapixel photo. On the processor (no graphics card, or **Use GPU for AI tools** off) several tiles are worked on at once, one per processor core, as many as the memory left allows: about 2 to 3 minutes for 24 megapixels on a 4-core computer, longer on a laptop running on battery. The window stays usable meanwhile, and **Cancel** stops it. The result is the same on every computer.

The model

AI denoise uses one network, trained on the noise of Lumix RAW files as Lab develops them; it works well on other cameras' files too. Dark areas come out clean and without blotches. The result is smooth: flat areas can look a little plastic, and strongly saturated yellows lose a little saturation.

If it looks too smooth, lower **Strength**: it blends the original photo back in, and with it some of its natural grain (around 70–85 often looks right).

A denoised layer made with an earlier model (before version 2.2) keeps its pixels and shows **Made with an earlier denoise model; Denoise again uses the new one**. Click **Denoise again** to redo it with the current model, in one undo step.

After the network, Lab gives the edges back: where the denoised photo has an edge, its own brightness detail is restored from the photo, and a light sharpening follows. Flat areas keep the denoised result, so the noise does not come back, and colours are not touched. Denoised layers made before version 2.2 keep their pixels; **Denoise again** makes them with the edge restore.

Every tile the network returns is checked. A tile that comes out wrong (stripes, a colour cast, values far out of range) is run again with its brightness range lifted for the network and put back afterwards (the network can go wrong on very dark areas), then on the processor. If it is still wrong, that part keeps the photo as it was, and a message says so when the denoise finishes. The edge restore and sharpening are never applied there.

Use GPU for AI tools, under the AI denoise button, is on by default. Untick it to run AI denoise, the AI masks, AI fill and Lens blur on the processor. This is slower but useful if your graphics driver gives wrong results. The setting is kept on this computer.

10 RAW files and the Library LAB

Camera RAW

Open a RAW file like any photo. The camera's own preview shows at once while Lab develops the RAW; after a second or two (more on a small laptop) the developed photo replaces it, at full size, with the camera's white balance and its colour matrix. The preview is only a picture: editing starts when the developed photo is in. The photo layer then has a **RAW develop** panel in the Layers tab:

- **Rendering**: **Accurate** (the default) keeps the scene's colours and the detail in bright lights, and matches Camera Raw's brightness. Coloured lights (LEDs, tail lights, signs) roll off softly to white, as on film, instead of ending in a flat white disc with a hard coloured rim. The core of a light the sensor clipped stays pure white: nothing is invented there. Every RAW you open now is developed this way. A photo developed with an older rendering shows its name here instead (**Camera-like (before 2.2)**, the look matched to the camera's own JPEG, or **Earlier Accurate**) and keeps it until you update it (below).
- **White balance**: as shot, or Temperature (Kelvin) and Tint.
- **Exposure** and **Highlight recovery**. With Accurate, Highlight recovery starts at 100, so everything the sensor recorded fits under white.
- **Highlight roll-off** (Accurate): how lights fade to white. **Tight** (0) starts close to the clip and keeps the glow near the light; **Standard** (50, the default) is matched to Camera Raw; **Filmic** (100) starts earlier and spreads the transition wider. Everything darker than where the roll-off starts is unchanged.
- **Warm highlights** (Accurate, 0 by default): the colour a light gives up on its way to white turns warm, as on film. A blue light stays blue-white, a red one turns orange-white instead of pink-white; white lights stay white.
- **Colour noise** (Accurate, 25 by default): removes the colour speckle of high-ISO photos and keeps real colour edges. It changes colour only, never brightness, so the grain stays (use AI denoise for that). 0 turns it off.
- **Lens corrections**: distortion, colour fringes and vignetting. These come from the camera's own data (Panasonic) or the Lensfun database. If there is no data for your lens, the panel says so.

While you drag a slider, a quick preview follows it several times a second; when you let go, the full-size develop replaces it (about a second on a 4-core computer). A new develop replaces the layer as one undo step. If you have already painted on that layer, Lab asks first.

Projects saved before version 2.2 open exactly as they were saved, and they keep the Camera-like look when you change a setting. Camera-like can no longer be picked for a photo. RAW layers from an earlier 2.2 test version keep their first Accurate look, and photos developed with an earlier Accurate keep theirs (where a bright blue light could get a violet ring). For all of these, the panel shows **Update to the new develop**. It develops the photo again with the current Accurate, as one undo step. **Copy edits** on a RAW photo also copies its develop settings, so pasting them onto other RAWs, in the editor or in the Library, gives them the same white balance, exposure and rendering.

Supported cameras and known limits

Lab reads RAW files with LibRaw 0.22.1: RW2 (Panasonic Lumix), ARW (Sony), CR3 and CR2 (Canon), NEF (Nikon), RAF (Fujifilm), 3FR and FFF (Hasselblad), DNG, ORF, PEF and more. Colour is tuned on Lumix

files first; other cameras use their colour matrix from LibRaw.

- **Nikon High Efficiency RAW (HE and HE*)** cannot be read yet. The Z 9, Z 8, Z f and Z6III can record it; Lossless Compressed, Compressed and Uncompressed NEFs from the same cameras open normally. Opening an HE file shows: **This Nikon file uses High Efficiency (HE or HE*) RAW compression, which Emulsion cannot read yet. Set the camera to Lossless Compressed RAW, or convert the file to DNG with Adobe's free DNG Converter first.** In the Library such a file still shows its preview, marked **Can't edit: Nikon HE**.
- Any other RAW that LibRaw cannot read shows **This RAW format is not supported yet** with the camera's name.
- **Convert and open.** For both, Lab can use **Adobe DNG Converter**, if you have installed it on this computer. It is free from Adobe; Emulsion never downloads, bundles or installs it. When Lab finds it (in its standard folder under Program Files, or wherever you pointed Lab to it with **Locate converter** once), the message comes with **Convert and open**. Lab converts a *copy* of the file to a DNG and opens that. The original is never changed, and the DNG keeps the camera's mosaic data (lossless compressed, not linear), so the RAW develop works on it exactly as on any other RAW. A Z 9 HE file takes a couple of seconds; Cancel stops it, and it stops on its own after two minutes. If the converter is not found, the message says **Adobe DNG Converter is free from Adobe: install it, then try again**, with **Locate converter**.
- The converter says nothing when it cannot read a camera (an older version and a newer camera, for example), so Lab only counts a conversion when the DNG is there and it can open it itself. Otherwise you see **Adobe DNG Converter could not convert this file (its version may not support this camera; update it)**.
- The same works from Recent, the Library (**Edit**, or select several and use **Convert N files**, which converts them one after another with a progress bar and Cancel) and the Edit Queue (files that could not be opened are offered together). A file dropped onto the window has no place on disk Lab can give the converter: open it with **Open** instead.
- Converted DNGs are kept in Lab's cache folder (not next to your photos), so opening the same file again is instant. They take at most 5 GB; the oldest go first. **About** shows how much they take and has **Clear converted files**. The photo layer says **Converted from NEF (DSC_9769.NEF) with Adobe DNG Converter**. A project keeps both the original's and the DNG's location. If the DNG was cleared, changing the develop asks to **Convert again**; without the converter it says so, and the layer always keeps its pixels.
- Batch export uses the converted DNG of a file when it is in the cache; files never converted are skipped and counted as before.
- A file that cannot be read never stops anything else: the photos already open stay as they are, an Edit Queue tab for it closes with the message, and batch export skips it and says how many were skipped and why.

The Library

Switch to **Library** in the header to browse and cull whole shoots. Add a folder or a memory card. Thumbnails appear at once, and your photos are never changed.

Key	In the Library
1 – 5, 0	Star rating, or clear it
6 7 8	Label Delete (red), Maybe, Keeper
Click, Ctrl +click, Shift +click	Select a photo, add or remove one, select a range
Ctrl + A / Esc	Select every photo shown / clear the selection
A	Add the selected photos to the Edit Queue, or take them out
Space	Full view (← → to move, Esc back)
Enter	Edit the photo
C	Compare two photos side by side

Editing several photos

Select photos with Ctrl +click or Shift +click. A bar appears over the grid with the number selected:

- **Edit Selected (n)** opens each photo in its own tab. The first opens right away. The others wait in their tabs ("Opening …") and load when you first click them, so a big batch doesn't fill your computer's memory. A photo that is already open is shown, not opened twice.
- **Add to Edit Queue** (A) puts photos aside to edit later, for example while you cull. Queued photos show a **Queued** badge. Press A again to take them out. In full view, A and the **Add to Edit Queue** button work on the photo shown.
- **Edit Queue (n)** lists the queued photos. Remove any with ×, or click **Open all** to open them in tabs. The queue is kept with your Library between sessions.

Up to 12 photos can be open at once (4 on a touch screen). If you open more, the rest stay in the Edit Queue and a message says how many. Close some tabs, then open the queue again.

- **Delete red** moves all photos labelled Delete to the Recycle Bin, so you can restore them from there.
- **RAW+JPG pairs** show as one photo marked RAW+JPG. Browsing uses the JPEG; editing and export use the RAW.
- **Sharpness** scores each frame relative to the sharpest in its burst. For panning shots it scores the car, not the blurred background. **Select soft** picks the weak frames.
- **Paste edits** (Ctrl + Shift + V) applies edits copied in the editor to every selected photo.
- Add captions, keywords, an event and a location in the **Info** panel, and search by them. **Collections** and **virtual copies** are supported too.
- **HDR** merges 2 to 9 bracketed exposures. **Panorama** stitches 2 to 9 frames, with **Crop to fill**. Each result opens as a new photo.
- **Batch export** writes the selected photos to a folder you pick. You set the format, quality, long-edge size, output sharpening, watermark and a file-name template such as `{name}` `{n}` `{date}` `{stars}` `{camera}`. Existing files are never replaced. A progress bar shows how far it has got; **Cancel** stops the batch, and the photos already written stay in the folder.

11 Exporting and saving projects

Export a finished photo

1. Click **Export**.
2. Pick the **format**: JPEG, PNG or WebP. LAB also offers TIFF. For JPEG and WebP, set the **quality**.
3. Pick the **size**: 100%, 75%, 50% or 25%.
4. Optionally add a **watermark**: text (by default "© year Artist") or your own logo, with position, size and opacity. A preview shows how it looks.
5. Choose the **metadata**: keep the camera info (EXIF) or not, keep the location (GPS) or not (off by default, safe for posting online), and fill in Artist and Copyright.
6. Click **Save**. The dialog shows the estimated file size before you save.

While a photo exports, a progress bar shows how far it has got, and **Cancel** stops it without writing a file. Large photos are rendered in bands, so even a 45 MP photo with many layers exports without running out of memory.

Exports are tagged sRGB with the profile embedded (see Colour and export).

	Lite	Lab
Largest size	2048 px on the long edge	Full size
Formats	JPEG, PNG, WebP	JPEG, PNG, WebP, TIFF (8-bit; None, LZW or ZIP compression; DPI)
Several open photos	One photo per export	Tick photos in the list (All / Only this one / None). They export with the same settings into a folder you pick, as name + "-edit", numbered rather than replaced.

Colour and export

Emulsion edits in **sRGB**, the colour space of the web, of most phones and of most monitors. Everything you open is turned into sRGB first, and everything you export is sRGB, with the profile embedded so every program shows the same colours.

Opening photos with other colour spaces

- **Photos with an embedded profile** (Adobe RGB, Display P3, ProPhoto and others, from Photoshop, Lightroom, phones) are converted to sRGB as they open. They look the same as in Photoshop, apart from colours that lie outside sRGB, which come out as the nearest colour sRGB can show.
- **Cameras set to Adobe RGB** write no profile into their JPEGs, only a mark in the camera info (EXIF). Emulsion reads that mark and converts the photo, so it no longer looks dull. The same goes for the camera previews inside RAW files, the Library's thumbnails and loupe, the JPEG of a RAW+JPG pair, merges, and a DNG whose previews are Adobe RGB. LAB's RAW develop does not depend on this camera setting: a RAW is always developed straight from the sensor data into sRGB.
- **A photo with no profile and no mark** is taken to be sRGB, as every browser does.

- **A damaged or unsupported profile**: a message says so, and the photo is shown as sRGB. Colours may be off; re-save the file from the program that made it.
- LAB opens **TIFF files**, 8 and 16 bit, with their profile applied in full 16-bit precision before the photo becomes 8-bit sRGB. TIFFs with JPEG compression, 32-bit floating point, CMYK or indexed colour are refused with a message saying what to change.

What exports carry

- Every JPEG, PNG, WebP and TIFF is **sRGB with the profile embedded**, and its camera info says sRGB too, even when the camera was set to Adobe RGB.
- A **PNG** of a photo with no transparency is saved as 8-bit RGB, which makes the file smaller with exactly the same pixels; a picture with transparent areas keeps its transparency.
- Adobe RGB, Display P3 and 16-bit output are not offered: the editor works in 8-bit sRGB, so they would add no colours or detail.

Tips

- For big brightness changes on a RAW photo (lifting a night shot by several stops), use **Exposure** in the RAW develop panel LAB: it works in floating point before the photo becomes 8-bit, so dark areas stay smooth. The same lift in Light & color on top can show steps (banding) in smooth dark areas.
- On a wide-gamut monitor, colours are right as long as Windows has the monitor's colour profile (Settings › Display › Color profile). Without it, everything on the screen, not just Emulsion, looks too saturated.

Save a project

An **.emulsion** project keeps everything: layers, masks, adjustments, applied presets and their Amount. Open it later to keep editing.

- **Save** or Ctrl+S. In Lab the next Ctrl+S writes to the same file. Ctrl+Shift+S is **Save as**.
- In Lite the project downloads as a file. Open it again with **Open**.
- A project made in Lab opens in Lite too. Lab-only layers such as Smooth skin are kept in the file but not shown in Lite.

Autosave and recovery

Emulsion keeps a recovery copy of unsaved work. If the app or browser closes unexpectedly, it offers to **recover** the work at the next start. Lab asks "Save changes?" when you close it with unsaved edits; in Lite the browser warns you.

History snapshots

In the **History** tab, name a snapshot to bookmark a state. **Restore** goes back to it as one undo step.

12 Keyboard shortcuts

Press ? in the app for the full list. Lite uses a different key where the browser keeps a shortcut for itself.

Action	Lab	Lite	
Files and view			
Open	Ctrl + O	Ctrl + O	
New blank canvas	Ctrl + N	Ctrl + Alt + N	
Save project / Save as	Ctrl + S / + Shift	Ctrl + S	Lite downloads
Next / previous photo	Ctrl + Tab / + Shift	Ctrl + Tab / + Shift	
Close photo	Ctrl + W	–	browser keeps it
Find an adjustment	Ctrl + F	Ctrl + F	
Before (hold)	\	\	
Clipping warnings	K	K	
Pan / zoom	Space +drag / scroll	same; pinch on touch	
All shortcuts	?	?	
Edit			
Undo / Redo	Ctrl + Z / Ctrl + Shift + Z	same	Ctrl + Y also redoes
Copy / paste edits	Ctrl + Shift + C / V	same	
Duplicate layer (no selection)	Ctrl + J	Ctrl + J	
Reorder layers	drag a row	drag; touch: hold, then drag	Esc cancels
Select layers: add or remove / range	Ctrl +click / Shift +click a row	same	Layers list
Merge selected layers	Ctrl + E	Ctrl + E	two or more selected
Brush size	[ ]	[ ]	
Selection			
Select all / Deselect	Ctrl + A / Ctrl + D	same	Esc also deselects
Invert selection	Ctrl + Shift + I	same	

Action	Lab	Lite	
Add / Subtract / Intersect	Shift / Alt / Shift+Alt + drag	same	
Copy / cut to new layer	Ctrl+C or J / Ctrl+X	same	
Transform	Ctrl+T	Ctrl+Alt+T	Enter / Esc
Clear / content-aware remove	Delete / Shift+Delete	same	
Tools			
Move · Pan · Select · Lasso	V H M L	same	
Wand · Object · Crop	W O C	same	Crop: Enter, Esc, X
Brush · Eraser · Clone · Remove	B E S R	same	
Gradient · Picker	G I	same	
Heal · Liquify · Dodge & burn	J Q D	–	Lab tools
Heal: another source / pick	/ / Alt+click	–	
Library LAB			
Select / add or remove / range	click / Ctrl+click / Shift+click	–	
Select all / clear	Ctrl+A / Esc	–	
Edit Queue: add or remove	A	–	the selected photos
Stars · clear · labels	1–5 · 0 · 6 7 8	–	Delete, Maybe, Keeper
Full view · compare · edit	Space · C · Enter	–	
Paste edits onto the selected	Ctrl+Shift+V	–	

On a Mac keyboard, use Cmd wherever this table says Ctrl (Lite in Safari).

13 Lite and Lab compared

Feature	Emulsion Lite (free)	Emulsion Lab
Where it runs	Any modern browser, computer or phone, at lite.emulsionlab.io	Windows desktop app
Price	Free	One-time purchase, no subscription
Network	None besides loading the page and its presets	Only to activate or deactivate the license
Layers, masks, blend modes, all 13 shared adjustments, selections, Transform, crop and geometry	✓	✓
Drag to reorder layers	✓	✓
Clone, Remove, Gradient, Brush, Eraser	✓	✓
Built-in presets (431)	Applied as they are, at most 2 stacked	Editable, unlimited stacking, Reset to preset
Your own and imported .xmp presets	✓	✓
Projects (.emulsion), autosave and recovery	✓	**✓ plus Save to the same file and recent files**
Largest photo opened	36 MP (about 16 MP on iPhone/iPad)	250 MP
Export size	2048 px long edge	Full size
Export formats	JPEG, PNG, WebP	+ TIFF
Export several photos at once	–	✓
Camera RAW (with lens corrections)	–	✓
Library: culling, ratings, sharpness, batch export, HDR, panorama	–	✓
AI masks, Select subject, AI tracing, AI fill, Lens blur, AI denoise	–	✓
Heal, Liquify, Dodge & burn, Frequency separation, Smooth skin, Halation	–	✓

14 Troubleshooting

A tool or button is greyed out with “Included in Emulsion Lab”

You are in Emulsion Lite (lite.emulsionlab.io), and that feature is part of Lab. The link explains what Lab adds.

My photo opened smaller than the original (Lite)

Lite reduces photos above 36 megapixels (about 16 on iPhone and iPad) so the browser can handle them. Lab opens them at full size.

My export is only 2048 pixels wide (Lite)

That is Lite’s export limit. Lab exports at full size.

I can’t apply a third preset (Lite)

Lite stacks at most 2 presets. Remove one in the Applied list, or use **Replace**.

My imported presets disappeared (Lite)

In a private or incognito window the browser forgets saved presets when the window closes. Emulsion warns you when this happens. Use a normal window.

“Not enough memory to open …”

The photo needs more memory than Emulsion has left on this computer (see Photo size limits). Close a photo you are not using, or another program, then open it again. AI denoise on the processor says the same when it has no room to start.

Sharpening or grain looks different when zoomed in

The editor previews at a reduced size for speed. Zoom to 100% to judge fine detail; the export is always rendered at full quality.

I moved a layer and its mask stayed behind

Masks belong to the document, not the layer, so they don’t follow moves. Move the layer back, or repaint the mask.

An adjustment went onto a new layer instead of the one I picked

A selection was active (or the layer was limited to a selection and none was active). See Adjusting only inside a selection. To edit a layer exactly as it is, click it in the Layers list.

A layer from a Lab project is invisible in Lite

Smooth skin layers are kept in the project but are not drawn in Lite. Open the project in Lab to see them.

Some settings of an imported .xmp preset are missing

Texture and the parametric curve sliders are not imported. Emulsion tells you which settings it skipped.

AI tools are slow LAB

Without a WebGPU-capable graphics card and driver, the AI runs on the processor. That works but is slower. AI denoise shows progress and can be cancelled. The same happens when **Use GPU for AI tools** (AI denoise section of the Layers tab) is off.

Windows says “Windows protected your PC” when installing LAB

Click **More info**, then **Run anyway**.

Activation messages LAB

Message	What to do
This license key is not valid.	Check it against your purchase email. Copy and paste it rather than typing.
…already activated on as many computers as it allows.	On the old computer, open **About > License** and click **Deactivate this computer**, then try again.
This key is for a different product, not Emulsion Lab.	Use the key from your Emulsion Lab purchase.
This license key has expired / has been disabled.	Email contact@emulsionlab.io with your purchase email.
Could not reach the license server.	Check your internet connection and try again. You only need it for activation.

Something else went wrong

Lab and Lite keep a recovery copy of unsaved work. Restart the app (or reload the page) and accept **Recover**. Click the logo for **About**, which shows the version you are using. Still stuck? Email contact@emulsionlab.io with your purchase email (Lab) and the version from About.